SWEEP rule on the KD-Toric code

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1 SWEEP Rule Decoder on the Toric

The decoder works as follows: each vertex v looks for a suggestion $\hat{v}$ on $\uparrow(v) \cap lk_2$ that matches σ_v^+. If there are more than one, then pick the one which has the minimal weight.

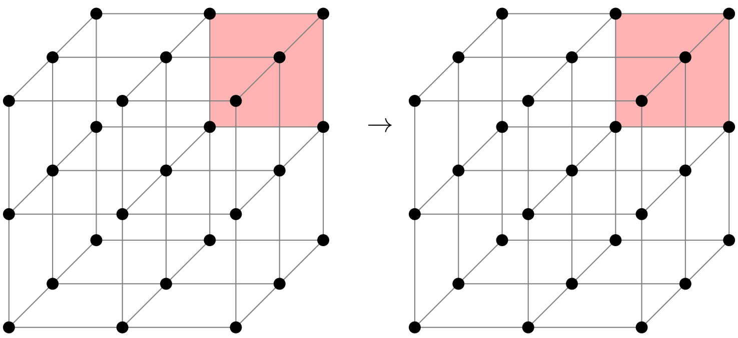

Figure 1: SWEEP rule

2 Span Expansion

Define the potentials functions $L_i : \mathbb{R}^d \to \mathbb{R}$ as follow[1]:

$$L_i(\mathbf{x}) = \begin{cases} \mathbf{x}_i & i < d+1 \\ -\langle \mathbf{1}, \mathbf{x} \rangle & i = d+1 \end{cases}$$

The **Size** of a subset S is defined as:

$$\mathbf{Size}(S) = \sum_i \max_{\mathbf{x} \in S} L_i(\mathbf{x})$$

Numerically, we show that the span property holds for the checks. Namely, let e be an unsatisfied check at time t, and let H_e be the unsatisfied checks at time $t-1$. Then there is a set $H'_e \subset H_e$ such that:

1. For any $i < d+1$ there exists $e' \in H'_e$ such $L_i(e') \geq L_i(e)$.
2. There exists $e' \in H'_e$ such that $L_{d+1}(e') \leq L_{d+1}(e) + 1$.

Proof. Consider a vertex v which at time t adjoins to an unsatisfied edge e, let u, w be vertices which have a positive faces that adjoin to e.

[1] Notice that we use opposite signs relative to Berman and Peter Gacs, yet we agree with Perskil.

1. If $\hat{u}|_e \neq \emptyset$, then the parallel edge to e going out from u belongs to H_e. Denote it by e_u, taking e_u satisfies at least two (three in 4D) of the first d inequalities and the last $d+1$th inequality. Denote by $x \in [d]$ the index of the inequality which isn't satisfied by e_u. So it remains to prove the existence of $e' \in H_e$ for which $L_x(e') \geq L_x(e)$.

   Consider the assignment of bits at time $t-1$ over the complex, and consider the graph $G = (\mathcal{F}, E)$ induced by associating vertices with any face that is set to 1 by the assignment. Two vertices are connected if their origin faces share an edge. Let $K \subset \mathcal{F}$ be a connected component of faces that contains the faces supported on u.

   We say that K is d far from u if:

   $$\max_{f \in K} |f_x - u_x| \leq d$$

   Now, we will prove by induction on d that there is a face $f \in K$ with $f_x \geq v_x$ such one of its edge is not satisfied.

   (a) **Base.** The edge from v which point at the $\hat{z}$ direction must not be satisfied.

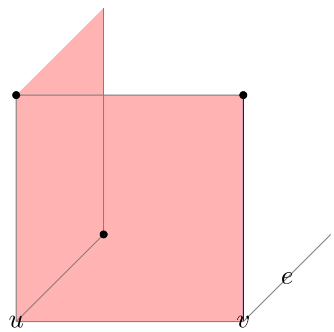

Figure 2: The induction base: In red, we have the faces on the support of u. The blue edge has the same x-coordinate as e. Any attempt to zero out the syndrome on the edge results in adding a face that is at least 2-far from u. Any configuration which is at most 1-far from u has to have that local view, yet might be connected to more faces which don't adjoin the blue edge.

   (b) **Assumption.** Assume that for any K which is at most $d-1$ far from u there is a face, with adjoins edge that is both has a x-coordinate that is greater than u_x and is not satisfied.

   (c) **Induction Step.** Consider a connected component K that is d-far from u. Notice that if by substitute a co-boundary (stabilizer) from K we get K' that is at most d' far from u for some $d' < d$ then we done. That because: $\partial_2 K' = \partial_2(K+z) = \partial_2 K$ for some $z \in \ker \partial_2$.

   On the other hand, **[COMMENT]** Finish that.

2. Else, if for any $u \in \downarrow(u) \cap lk_1$ it holds that $\hat{u}|_e = \emptyset$, then it must be that $e \in H_e$ because $\hat{v}$ can only decrease the syndrome on v. Moreover, it follows that $\hat{v} = \emptyset$ (otherwise, the suggestion would zero out the syndrome on $\uparrow(v) \cap lk_2$). Thus, $\sigma_v \not\subset \uparrow(v) \cap lk_2 \Rightarrow$ there is also an edge $e' \in H_e$ where $e' \in \downarrow(v)$. In that case, take $e \in H_e$ as the edge satisfies the first d equalities and e' for satisfying the $d+1$th inequality.

□

3 Investigation

[COMMENT] To do: understands how the investigation should look like. Plot an example to such tree. The model goes as follow, at each even iteration we toss a noise, then in each odd iteration we perform a single SWEEP rule iteration. That allows to track over the history of the errors.

We say that a check $e \in$ Error, if there is a face f that adjoin to e and belongs to the Error.

Definition 3.1 (Investigation)**.** For any unsatisfied check e at time t, we define V (the set of checks which participated in 'misleading' of e), and two sets of edges on V, Arrows and Forks via the following algorithm:

```
let V = {e}, Arrows = ∅,
  Forks = ∅
let Q ← queue {e}
while  Q is not empty do
    e' ← dequeue Q
    if  e' ∉ Error then
        e_1, .., e_l ← Excuse( e'
          )
        Insert {e_i, e'} to
          Arrows
        Insert {e_i, e_j} to
          Forks
    end
end
```

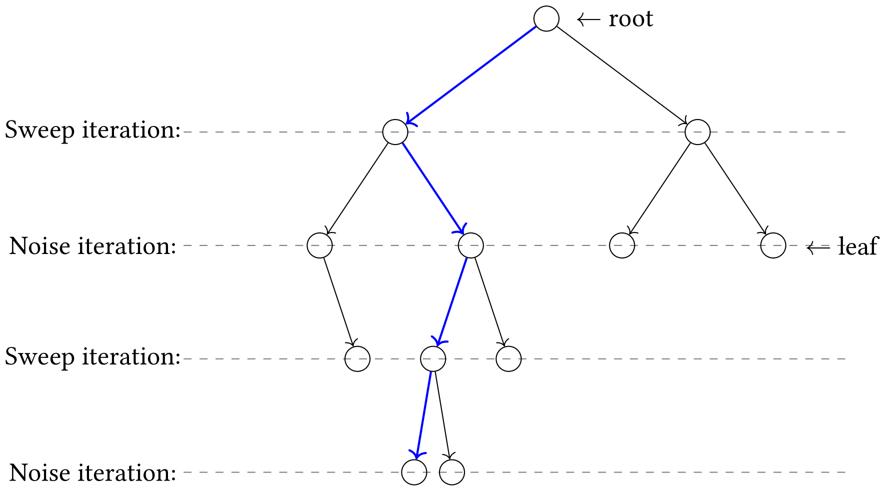

Figure 3: blabla

Δ[01History(1) Δ[01Time(1) Δ[01Excuse(1)

Definition 3.2 (History, slice.)**.** G_u is the subgraph of $\langle V, \text{Arrows}\rangle$ induced by $\{v \in V : v \leq u\}$. u is the connecting connected component of G_u containing u. A slice is an equivalence class of the relation $u \equiv v$ iff $u = v$. Notice that if u and v are in the same slice K, then $u = v$. Thus we can write K and K without ambiguity.

Claim 3.1. If for two different slices K_1 and K_2 we have that $K_1) = K_2)$, then $K_1)$ and $K_2)$ are disjoint.

Proof. Directly from definitions. □

Definition 3.3 (The graph of slice)**.** For a slice K we define the graph $G_K = \langle V_K, E_K\rangle$ as follows:

1. $V_K = \{\text{slice } R : R \subset K \text{ and } R = K - 1\}$
2. $\{R, S\} \in E_K$ iff $R, S \in V_K$ and for some $\{v, w\} \in$ Forks we have $v \in R$ and $w \in S$.

Claim 3.2. G_K is connected.

Proof. Assume R and S are from V_K and that $v \in R, w \in S$. Since $v, w \in K$, there exists a path from v to w in K consisting of arrows. Select this path in such a way that it contains the minimal number of points x such that $x = K$. By induction on the number k of such points we prove that R and S are in the same connected component of G_K.

1. Base. $k = 0$, In this case, the path from v to w lies within $G_v (= G_w)$. Hence $v = w$ which means that the slices R and S are the identical.
2. Induction step. There exists y on the path from v to w such that $y = K$. Let x be the point which precedes y on the path, and z the point that follows y. Since the values of **Time** at the two ends of an arrow differ by 1, and since y is minimal, we know that:

(a) $y = \langle x, z \rangle$ or $y = \langle z, x \rangle$, thus $\{x, z\}$ is a fork.

(b) The slices of x and z, say T and U, are both members of V_K.

By the induction hypothesis, R and T are in the same connected component of G_K, and so are U and S. On the other hand, $\{T, U\} \in E_K$.

$\square$